CODING AGENTS

A practical intro for engineers new to coding agents

Start simple.

Scale smart.

slides and code

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Spring Developer Advocate · Broadcom

HELLO 👋

About me.

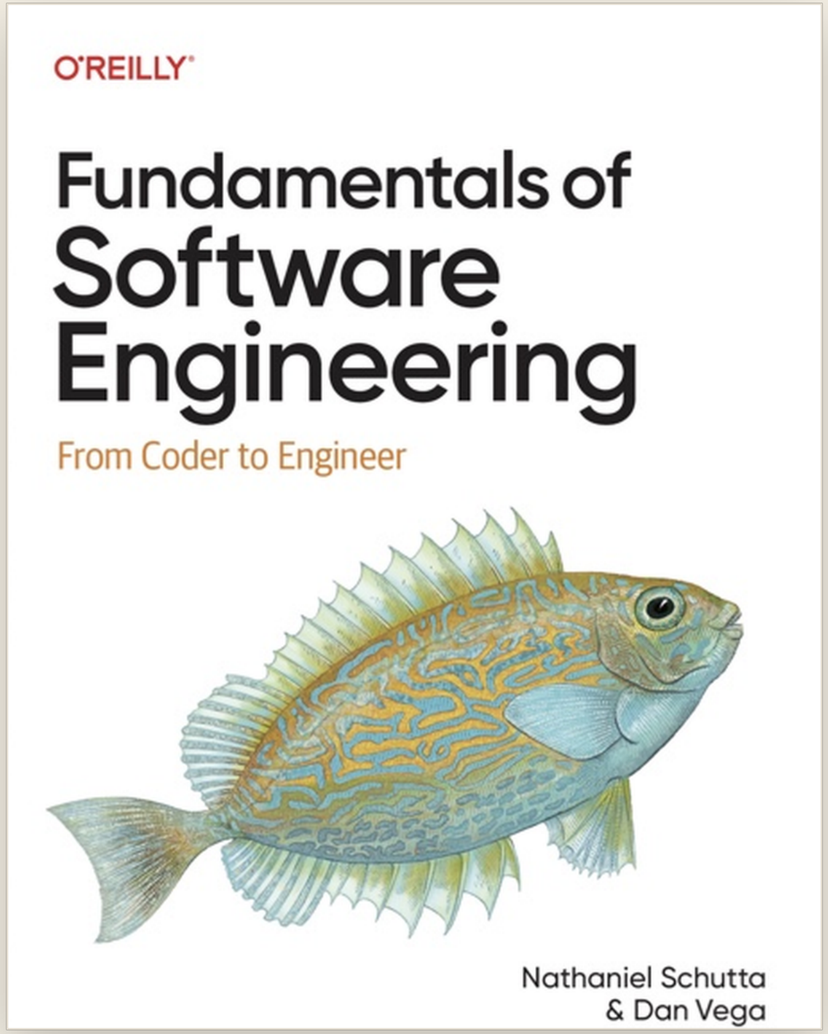
Spring Developer Advocate at Broadcom

Java Champion

Co-author of Fundamentals of Software Engineering

25 years as a software engineer

danvega.dev



One quick disclaimer.

If you're already running five terminals with custom agents and skills...

This talk probably isn't for you. Go grab another coffee. We'll be in the fundamentals for most of this talk.

CODING AGENTS

One tool today. Most of it translates to your favorite.

Codex Gemini CLI Copilot Cursor Windsurf Grok Build Kiro

I want to describe a feeling I get these days.



Morgan Lee

@shipwright_dev · 2m

five agents in five tabs, numbered. a notification pings me when one needs an answer. I barely touch the keyboard now

12 8 46 15



Kay Greenfield

@greenfield_kay · 7m

reviewed three PRs from the gym this morning. my agents opened four more while I was in the shower

17 23 89 31



Four Terminals

@fourterminals · 11m ·

if you're not orchestrating subagents in 2026, you're already obsolete

12 31 76 18



Cronjob Carla

@cronjob_carla · 18m ·

set up a routine that triages our backlog at 6am. it closed two issues before standup and nobody noticed for a week

9 27 64 11



NullPointer Nate

@nullpointer_nate · 24m ·

20 years in this industry and I have never felt this far behind

---

18 42 97 23

Social media driven development.

QUICK SHOW OF HANDS…

Who's feeling overwhelmed?

It's okay. This is a safe space.

Things we're supposed to learn in coding agents.

`CLAUDE.md` **Hooks** **Subagents** **MCP** **Worktrees** **Plan Mode** **Skills**

Agents `/clear` `SKILL.md` **Memory** `/init` **Plugins** **Agent Teams**

Things we're supposed to learn in coding agents.

CLAUDE.md
Hooks
Subagents
MCP
Worktrees
Plan Mode
Skills
Agents
/clear
SKILL.md
Memory
/init
Plugins
Agent Teams
Harness
Checkpoints
PostToolUse
Status Line
/context
Sonnet
Permissions
Routines
Tokens
/goal
Compaction
PreToolUse
Marketplace
/resume

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Marketplace
/resume
SessionStart
Auto Mode
Context Window
Side Chat
Background Sessions
Slash Commands
Headless Mode
Opus
Dispatch
Auto-Accept Edits
Cowork
Effort Levels
Haiku
.claude/
Computer Use

...and that's just this week

You are not behind.
You don't need to learn all of this.
You need the building blocks.

Scale later. We're all on our own journey.

TODAY

What we'll cover.

01 **The fundamentals** — agent · harness · token · context window

02 **Getting started** — three surfaces · permissions · plan mode

03 **Context is the job** — CLAUDE.md · skills · MCP

04 **Scaling up** — subagents · hooks · goals · worktrees

SECTION 01 / 04

The fundamentals.

The vocabulary everything else hangs on.

01 **The fundamentals**

02 Getting started

03 Context is the job

04 Scaling up

FUNDAMENTALS · 01

Agent.

FUNDAMENTALS · 02

Harness.

HARNESS

The machinery that builds and runs the loop.

Claude Code is a harness.

So are Codex, Cursor, Gemini CLI, and Copilot's agent mode.

Same model, different harness, **different results.**

harness · **the car**

model · **the driver**

you · the destination

↓

decide → **use a tool** → **observe**

↻ repeats until the goal is done

tools	**context**	**safety rails**	**orchestration**
steering and pedals read, edit, run	windshield and mirrors what the model sees	brakes and seatbelts sandbox, approvals	cruise control retries, loops

Agent = model + harness. You pick the destination.

FUNDAMENTALS · 03

Token.

TOKEN

You don't pay for words. You pay for tokens.

`Ref` `actor` `the` `payment` `service` `and` `run` `the` `tests` = 9

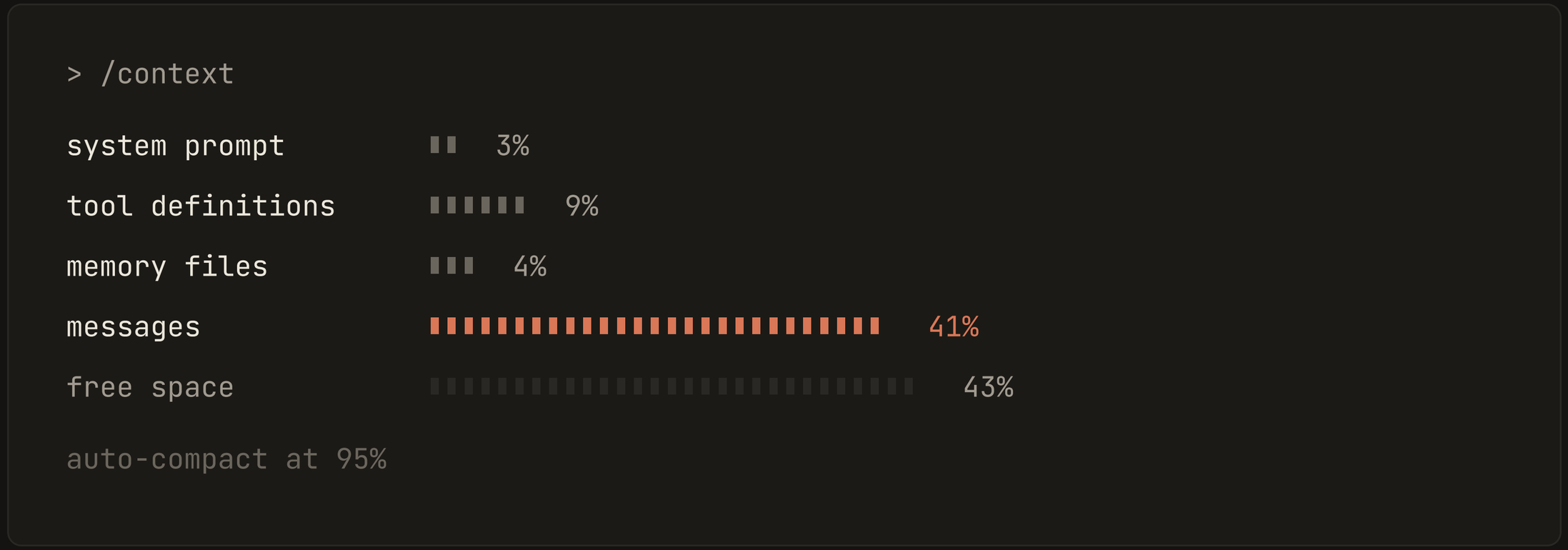
Roughly three quarters of a word. Claude reads, writes, and bills in them.

FUNDAMENTALS · 04

Context window.

CONTEXT WINDOW

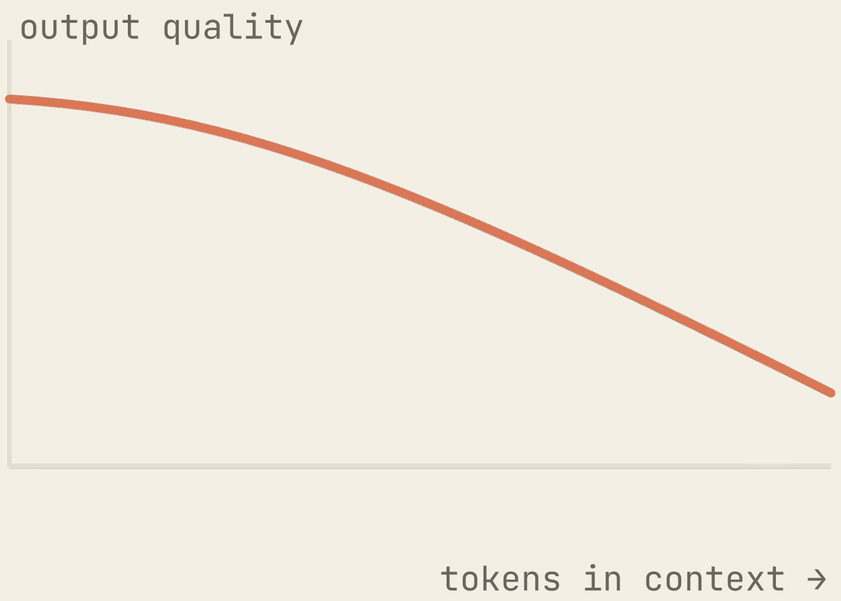
Working memory for one session.



Everything Claude is holding in its head right now. Big, but finite.

THE CATCH

Context rot.



More context is not better context.

18 frontier models tested. All 18 got worse as the window filled.

Source: Chroma, "Context Rot" (2025)

SECTION 02 / 04

Getting started.

Three surfaces, permissions, plan mode.

01 The fundamentals

02 **Getting started**

03 Context is the job

04 Scaling up

THREE WAYS IN

Same engine underneath.

`terminal`

fast, scriptable

`desktop`

parallel sessions

`web`

runs while you're away

BEFORE YOU START

Two ways to pay.

$/month

Subscription. Flat, predictable.

Pro, Max, or a Team seat. One pool across chat and Code.

$/token

API. Metered, no ceiling.

Great for scripts and CI. Pricey for daily coding.

A stray `ANTHROPIC_API_KEY` silently moves you to the right column. `/status` shows your route.

DEMO · 01

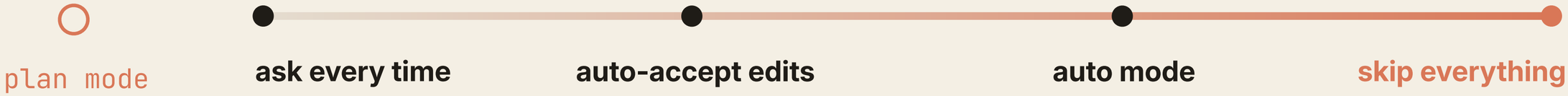
A first session.

Open a project. Run claude. Ask for one small thing. Watch the loop.

→ jump to claude code

TRUST, BUT VERIFY

Permissions.



THE ONE FOLDER TO KNOW

.claude/

Everything custom about your Claude Code setup lives here. Memory. Commands. Agents. Skills. Hooks.

```
.claude/
├── commands/your custom slash commands
├── agents/subagent configs (markdown)
├── skills/SKILL.md folders
├── hooks/lifecycle shell scripts
├── settings.json project settings
└── settings.local.json your overrides (gitignored)
CLAUDE.md project memory (root of repo)
```

INSIDE .CLAUDE

settings.json

One JSON file controls permissions, environment, and team-wide defaults. Checked into the repo, it travels with the project.

.claude/settings.json

```
{
  "model": "sonnet",
  "permissions": {
    "allow": ["Bash(./mvnw *)", "Bash(git diff *)",
              "Read"],
    "deny": ["Bash(rm -rf *)", "Bash(mvn deploy *)",
             "Read(./.env)"],
    "ask": ["Bash(git commit *)", "Bash(git push *)"]
  },
  "env": {
    "SPRING_PROFILES_ACTIVE": "test"
  }
}
```

MAYBE MY FAVORITE FEATURE

Plan mode.

Hit ⇧ Tab

until plan mode. Claude goes read-only.

The difference between "oh no" and "yes, do exactly that."

01 **Explore the codebase**

02 **Ask clarifying questions**

03 **Write a markdown plan**

04 **You approve · it executes**

NO PLAN MODE?

Prompt it. Or make it a skill.

PROMPT IT

Ask for the plan first.

"Explore the code. Ask me questions. Write a plan. Don't edit anything until I approve it."

SKILL IT

Write it down once.

Save that prompt as a skill. Any agent that supports skills can use it.

More on skills in a bit.

DEMO · 02

Permissions and plan mode.

A new project, from first prompt to trusted setup.

→ jump to claude code

SECTION 03 / 04

Context is the job.

Steering Claude with what you already know.

01 The fundamentals

02 Getting started

03 **Context is the job**

04 Scaling up

This is where you can take a coding agent from a nice parlor trick to an irreplaceable teammate.

The model is only as good as the context you give it.

THREE GAPS, THREE FIXES

Three ways the model comes up short.

GAP 01

CLAUDE.md

What it should always know about this project.

GAP 02

Skills

What it should know how to do.

GAP 03

MCP

What it cannot see from where it sits.

PROJECT MEMORY

CLAUDE.md

A markdown file at the root of your repo.
Claude reads it every session.

20 minutes on a good one makes every session sharper.

CLAUDE.md

```
# spring-petclinic
Java 25 · Spring Boot 4.1 · PostgreSQL · Maven
## Conventions
- Constructor injection only
- Records for DTOs
- Tests use Testcontainers, not H2
## Run it
$ ./mvnw spring-boot:run
## Gotchas
- Migrations live in db/migration, not resources/
```

Have an `AGENTS.md`? Claude reads it when there's no `CLAUDE.md`.

PROJECT MEMORY · THE CRAFT

Write one that actually helps.

VAGUE · DON'T

"Write good code."

"Follow best practices."

"Format things properly."

"Test your changes."

Claude already tries to do all of this. Zero signal.

SPECIFIC · DO

"Use 2-space indent, no semicolons."

"Constructor injection only."

"DTOs are records in model/."

"Run ./mvnw verify before done."

Concrete rules Claude can't guess. Real signal.

Structure it.
Headings and bullets, not prose.

Keep it tight.
Every line costs context.

Revisit it.
Fix the file, not just the chat.

KEEP IT LEAN

Progressive disclosure.

Point at the details instead of pasting them. Claude reads a file only when the task needs it.

CLAUDE.md

```
# Payments API
Constructor injection only.
Run ./mvnw verify before done.
Testing conventions: @docs/testing.md
DB migrations: @docs/migrations.md
Release process: @docs/release.md
```

Loaded up front: 8 lines. Loaded on demand: everything else.

GAP 02 · WHAT IT SHOULD KNOW HOW TO DO

A skill is a folder with a `SKILL.md`.

Expertise packaged once, pulled in on demand. Not crammed into every prompt.

SKILLS

What one looks like.

The description is the trigger. Claude reads it to decide when the skill applies.

The body is the playbook. Steps, conventions, commands.

It's just markdown. In the repo, reviewed and versioned like code.

.claude/skills/readme/SKILL.md

```
---
name: readme
description: Write, rewrite, or review
a README so it serves its actual
reader. Use whenever the task is
creating or improving a README.
---
A README has one job: get the reader
to what they came for in under a minute.
## Ground rules
- Open with one plain sentence saying
what the project is and who it's for
- Verify every command against the repo
- No hype. Show a code sample instead.
```

SKILLS

If you explain it twice, make it a skill.

`new-endpoint`	scaffold a REST endpoint the way this team does it
`db-migration`	write a Flyway migration, naming and all
`code-review`	review a diff against the team's checklist
`release-notes`	turn merged PRs into a changelog, in house style

SKILLS IN THE WILD

Four jobs, twelve skills.

VIDEO

video-edit
Raw footage to Premiere.

video-rough-cut
Cut dead air, fillers, and retakes.

video-motion-graphics
Graphics timed to what I say.

WRITING

blog-new-post
A first draft in my voice.

blog-seo-optimize
Find what developers search for.

newsletter-publish
Beehiiv edition to my site.

SHIP IT

shipit-shape
Find the smallest useful version.

shipit-mvp
PRODUCT.md before any code.

shipit-feature
Build the next feature and verify it.

PODCAST

spring-office-hours-edit
Raw stream to podcast mp3.

spring-office-hours-prep
Audio and show notes in one go.

podcast-appearance
Log my guest appearances.

Type the same prompt twice? Make it a skill.

GAP 03 · WHAT IT CANNOT SEE FROM WHERE IT SITS

MCP is a standard plug for outside tools.

One protocol. Any tool that speaks it, Claude can see and use.

MCP

What connecting looks like.

One command to connect. The server does the rest.

Claude discovers the tools. The server advertises what it can do.

You stay in charge. Tool use runs through the same permissions.

terminal

```
$ claude mcp add --transport http \
    dvaas https://mcp.danvega.dev/mcp
✓ Connected to dvaas
✓ 21 tools discovered:
youtube · blog · speaking ·
newsletter · podcasts
```

MCP

Wherever Claude is guessing, plug in the source.

`github`
read the real issue, comment on the real PR

`postgres`
query the actual schema, not a guess at it

`chrome-devtools`
drive a real browser: live DOM, console, network

`figma`
build from the design file, not a screenshot

`cloudflare`
manage workers, DNS, and logs on your real account

`context7`
current docs for your framework version, not training data

CONTEXT IS THE JOB

Skills versus MCP.

SKILLS

Teach Claude how.

Markdown playbooks for repeatable work. Static. Can't reach outside the repo.

```
"turn a newsletter into a site post"
```

MCP

Give Claude tools.

Live connections to tickets, databases, APIs. Capabilities. They don't teach taste.

```
"fetch the edition from Beehiiv"
```

TOGETHER MCP fetches the newsletter. The skill publishes it my way.

DEMO · 03

Skills and MCP.

Teach it a skill, then plug in `mcp.danvega.dev`.

→ jump to claude code

SECTION 04 / 04

Scaling up.

One session to many. Watching to delegating.

01 The fundamentals

02 Getting started

03 Context is the job

04 **Scaling up**

SCALING UP

An agent is a file.

.claude/agents/code-reviewer.md

```
---
name: code-reviewer
description: Reviews code for quality
tools: Read, Glob, Grep
model: haiku
---
You are a code reviewer. Give specific,
actionable feedback.
```

Its own prompt
you decide how it behaves

Only these tools
read only, in this case

A cheaper model
haiku for mechanical work

Its own context window
the whole reason to use one

It reads 40 files. You get back a paragraph.

DELEGATING SAFELY

Hooks. Hard rules, not vibes.

CLAUDE.md

Advises.

The model reads it. The model can drift.

`"Use BigDecimal for money."`

Claude usually listens.

HOOKS

Enforce.

A script that runs before or after Claude acts. Every time, no judgment call.

`Block any edit that stores money in a double.`

Claude can't get past it.

The more you delegate, the more you want rules that cannot be talked out of.

WORK THAT REPEATS

Routines.

Claude works while your laptop is closed.

	WHEN		CLAUDE
`schedule`	**Every weekday, 8am**	→	Triage new issues and label them.
`api`	**CI goes red**	→	Read the log and open a fix PR.
`webhook`	**A PR opens**	→	Review it and leave comments.

FROM WATCHING TO DELEGATING

/goal

Set the outcome, not the task. Claude keeps working, turn after turn, until it gets there.

```
/goal Every task in PLAN.md is marked complete,
and ./mvnw verify passes with output shown for each one.
A task is not complete until it has a unit test
and an integration test covering its behavior.
Update PLAN.md after each task.
Stop after 25 turns.
```

← THE OUTCOME

← HOW IT'S CHECKED

← WHEN TO STOP

KNOW THE DIFFERENCE

`/goal` is not spec driven development.

SPEC DRIVEN

A plan.

Every decision made up front. The agent builds to it.

/GOAL

A finish line.

Keep going until this is true. It doesn't decide what to build.

Don't reach for it when the task needs design decisions you haven't made yet.

MY TAKE

You may not need spec-driven development.

VIBE CODING

One prompt to rule them all.

Fun for prototypes. Without a plan, it falls apart in production.

SPEC DRIVEN

A spec for everything.

Spec Kit, Kiro, OpenSpec, BMad. Great for big teams. Overkill for one developer.

My middle ground: one PRODUCT.md and my own process, written as skills.

RUNNING MORE THAN ONE

Two agents, one repo, one mess.

Git worktrees give each branch its own folder. Same history, separate working files.

FROM CLAUDE

```
claude --worktree feature-auth
```

OR BY HAND

```
git worktree add ../api-auth auth-fix
git worktree remove ../api-auth
```

One .git history, shared across all of them.

Separate files, so builds and tests don't collide.

No stashing, no branch switching mid task.

Cost: each folder needs its own install and build.

DEMO · 04

Agents.

Working with subagents

→ jump to claude code

Remember the wall? You know the ones that matter.

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Opus
Dispatch
Auto-Accept Edits
Cowork
Effort Levels
Haiku
.claude/
Computer Use

You are not behind. Start simple, scale smart. **We're all on our own journey.**

SAVE YOURSELF A FEW WEEKS

Mistakes I made so you don't have to.

01 **Vague one-line prompts.** "Fix the bug." Cool. Which one? Where? What did you try? Be specific.

02 **Skipping plan mode for real work.** Five minutes of planning saves an hour of "what did you just do?"

03 **Never running /clear.** Stale context bleeds into new tasks. Wipe between unrelated jobs.

04 **No CLAUDE.md.** Re-explaining your stack every session. Twenty minutes, write it once.

05 **Trusting the first answer.** Read the diff. Run the tests. Push back. Claude likes the pushback.

06 **Reaching for Skills/MCP/Agents too early.** First learn what Claude does well bare. Then add scaffolding.



github.com/danvega/claude-code-start-simple

the code from this talk

That's it.

Thanks. Questions?

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